



▲ Yellowfin tuna (*Thunnus albacares*) is a fast-swimming schooling fish that is commercially important worldwide.

► Native to coral reefs of the Pacific Ocean, the brightly colored red lionfish (*Pterois volitans*) can inject venom through its spines, causing a severe and painful reaction in humans.



▲ The sea horse has a highly modified body form, as exemplified by *Hippocampus ramulosus*, shown above. Sea horses are unusual among animals in that the male carries the young during their embryonic development.



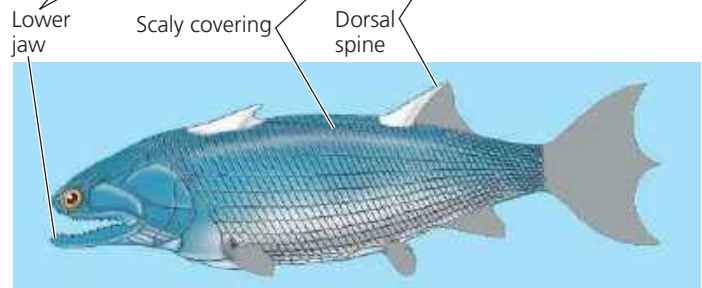
▲ The fine-spotted moray eel (*Gymnothorax dovii*) is a predator that ambushes prey from crevices in its coral reef habitat.

#### ▲ **Figure 34.17 Ray-finned fishes (Actinopterygii).**

➔ **Mastering Biology Video: Sea Horse Camouflage**

the Silurian period (**Figure 34.18**). The key derived character of lobe-fins is the presence of rod-shaped bones surrounded by a thick layer of muscle in their pectoral and pelvic fins. During the Devonian (419–359 million years ago), many lobe-fins lived in brackish waters, such as in coastal wetlands. There they may have used their lobed fins to help them move across logs or the muddy bottom (as do some living lobe-fins). Some Devonian lobe-fins were gigantic predators. It is not uncommon to find spike-shaped fossils of Devonian lobe-fin teeth as big as your thumb.

By the end of the Devonian period, lobe-fin diversity was dwindling, and today only three lineages survive. One lineage,



#### ▲ **Figure 34.18 A reconstruction of an ancient lobe-fin.**

Discovered in 2009, *Guiyu oneiros* is the earliest known lobe-fin, dating to 420 million years ago. The fossil of this species was nearly complete, allowing for an accurate reconstruction; regions shown in gray were missing from the fossil.

the coelacanths (Actinistia), was thought to have become extinct 75 million years ago. However, in 1938, fishermen caught a living coelacanth off the east coast of South Africa (**Figure 34.19**). Until the 1990s, all subsequent discoveries were near the Comoros Islands in the western Indian Ocean. Since 1999, coelacanths have also been found at various places along the eastern coast of Africa and in the eastern Indian Ocean, near Indonesia. The Indonesian population may represent a second species.

The second lineage of living lobe-fins, the lungfishes (Dipnoi), is represented today by six species in three genera, all of which are found in the Southern Hemisphere. Lungfishes arose in the ocean but today are found only in fresh water, generally in stagnant ponds and swamps. They surface to gulp air into lungs connected to their pharynx. Lungfishes also have gills, which are the main organs for gas exchange in Australian lungfishes. When ponds shrink during the dry season, some lungfishes can burrow into the mud and estivate (wait in a state of torpor; see Concept 40.4).

▼ **Figure 34.19 A coelacanth (*Latimeria*).** These lobe-fins were found living off the coasts of southern Africa and Indonesia.

